

Project 3: Descriptive Analysis on a Learning Platform

Dataset Overview

The dataset contains **29,604 entries** across 18 columns, capturing user-generated text and metadata from a Massive Open Online Course (MOOC) platform.

Key Attributes:

- **Numerical Metrics:** Posts are rated on scales of 1–7 for **Sentiment** (avg 4.18), **Confusion** (avg 3.91), and **Urgency** (avg 2.74).
- **Interaction Data:** Includes binary indicators for whether a post is an **Opinion**, **Question**, or **Answer**, as well as the number of **Reads** and **Up-counts**.
- **Course Metadata:** Data spans three primary course types: **Medicine** (the most frequent with 10,001 posts), **Humanities**, and **Education**.

Descriptive Observations

- **Engagement and Popularity**

The average post in this dataset has been read **410 times**, though the distribution is heavily skewed toward the lower end (median of 65 reads). A small number of posts achieve very high visibility, with the maximum reach recorded at **9,452 reads**. Interestingly, while up-votes are generally low (average 0.40), some highly valued contributions received up to **191 up-votes**.

- **Post Characteristics**

- **Anonymity:** Use of the anonymity feature is rare; only about **1.7%** of posts were made anonymously.
- **Post Types:** The data is divided between **Comments** (17,298 entries) and **CommentThreads** (12,287 entries). All 12,287 CommentThread entries are missing a `comment_thread_id`, suggesting that this ID is specifically used to link individual comments to their parent threads.
- **Content Distribution:** **Opinions** are the most common form of interaction, appearing in roughly **55%** of all posts. **Questions** and **Answers** are much less frequent, each appearing in approximately **20%** of the dataset.

Comparative Analysis

Sentiment, Confusion, and Urgency by course type

- **Medicine:** Posts show significant variance in urgency, with many maintaining consistently high ratings. These posts generally show lower levels of confusion or extreme sentiment.
- **Education:** This field averages higher scores for both sentiment and confusion, typically ranging between 4 and 5.
- **Humanities:** Engagement reflects moderate urgency, usually averaging between 2 and 4 on the 7-point scale.

Interaction Intent by Category

- **Content Trends:** Opinion-based posts in Education are nearly double the volume of other disciplines, though this field has the lowest frequency of Questions and Answers.
- **Discipline Engagement:** Medicine leads in average Questions and Answers per post, with Humanities engagement falling between the two extremes.

Post Type and Engagement Metrics

- **Engagement Skews:** Up-votes are higher on average for parent posts (CommentThreads) than for individual replies, Individual Comments receive significantly more "Reads" on average. This suggests that popular parent threads drive students to read through deep strings of subsequent replies.

Temporal and Behavioral Insights

- **Seasonal Trends:** Analysis shows a massive drop in post frequency at the start of the year, likely due to academic holiday periods.
- **Resolution Drivers:** There is a clear trend where posts marked with higher urgency have a statistically higher probability of being answered.
- **Anonymity and Sentiment:** Anonymity often correlates with negative outlooks; nearly 50% of anonymous posts express low sentiment, scoring between 0 and 2.